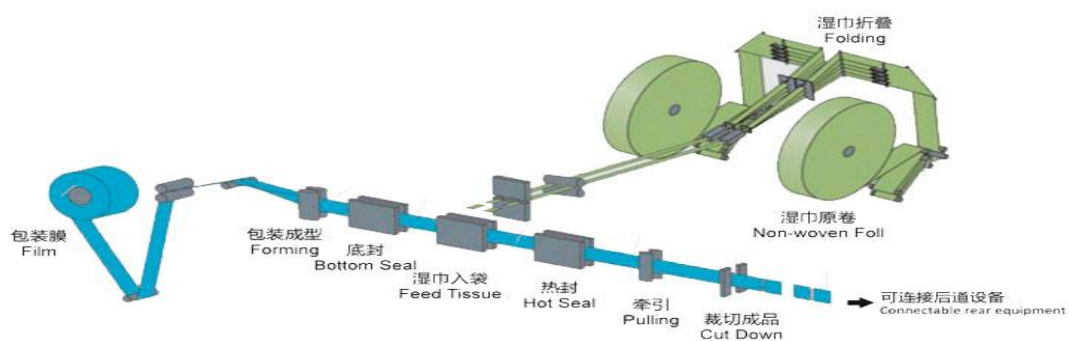


User Manual

VPD250 Automatic Horizontal Four-side Sealing Wet Tissue

Packaging Machine



Wenzhou Weipai Machinery Co., Ltd.

温州市伟牌机械有限公司

Address: No. 709 Zhengsong Avenue, Wanquan Town, Pingyang County, Wenzhou City

地址: 浙江省温州市平阳县万全镇郑宋大道 709 号

Website: www.packagingmachines.cn www.viroo.cn

Email: admin@viroo.cn viroo@machine.cc

Catalogue

Preface	2
1.Attention.....	2
1.1 Attentions.....	2
1.2. Marks in the Machine.....	3
2.Description of Machine.....	4
2.1 Summary.....	4
2.2 Main Technical Paramete.....	5
2.3 Structure.....	6
3. Installation of machine.....	7
3.1Requirement of floor.....	7
3.2Enviroment condition.....	7
3.3Test of eletricity supply.....	7
4. Machine setting and test.....	8
4.1.Operation instruction for control panel.....	8
4.2 Instructions for PLC operation.....	8
4.3 Temperature control	17
4.4 Installing packing film.....	18
4.5 Packing in fixed length mode.....	18
4.6 Packing in color code tracking mode.....	19
4.7 Adjusting packing speed.....	20
4.8 Tissue unwinding folding mechanism.....	20
5 Machine operation.....	22
5.1 Manual running.....	22
5.2 Automatic running	23
5.3 Stop.....	23
5.4 Operating steps.....	23
6. Maintenance and cleaning of the machine	24
6.1 External cleaning of the machine.	24
6.2 Inside cleaning of the machine.....	25
6.3. To lubricate the machine.....	25

7.General trouble shooting.....	25
8.Packing list.....	30

Preface

Please read the instruction manual carefully before operating the machine of our company for the first time, it will help you to correctly operate and maintain the machine so as to ensure safety, completely bring into play its performance, reduce troubles and prolong the use life.

1. Attention

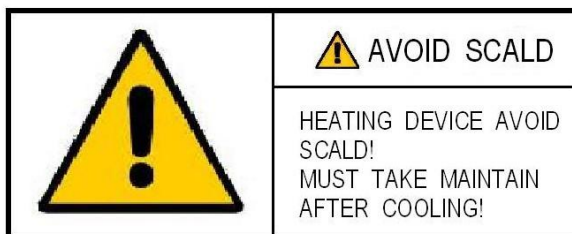
1.1 Attentions

- 1) Forbidden to start the machine before not clearly knowing the correct operation methods and safety rules.
- 2) Persons who were not trained and authorized can't use this machine.
- 3) You should read the manual carefully and know the whole content and points before using the machine.
- 4) You should first finish installation and necessary adjustment and then operate it.
- 5) You must be sure that there aren't any tools or things being put on the tables of machine before startup.
- 6) You can't touch the inside of the machine of electrical installation if not cutting off the electricity supply.
- 7) The operator cannot leave working range of machine in running.
- 8) Forbidden to touch the hot sealing gear, sealing mould and elements of the machine in running.
- 9) The electric professionals should finish inspecting and maintaining the electric control circuits.
- 10) You can't optionally reequip the machine or equip any nonpermissible devices, cutters or peripheral on the machine without permission of our company so as not to be in danger.
- 11) If you have any doubts, please ask our company or authorized franchisers, and not deal with them without authorization.
- 12) You shouldn't use the machine beyond the appointed condition.

Warning: Please obey the above safety items for you, others' and equipment's safety. Our company will not be responsible for suddenness or incidents caused by not complying with the above requirements.

Warning: Please obey the above safety items for you, others' and equipment's safety. Our company will not be responsible for suddenness or incidents caused by not complying with the above requirements.

1.2. Marks in the Machine





2. Discription of machine

2.1 summary

The machine is specially designed for packing wet tissue, alcohol pad, cleaning tissue, nursing tissue etc. It runs fast and stably. The machine can automatically complete: folding wet tissue, cutting, injecting chemical liquid, film slitting, bag making, heat sealing, counting, printing batch number, finished product output, and so on.

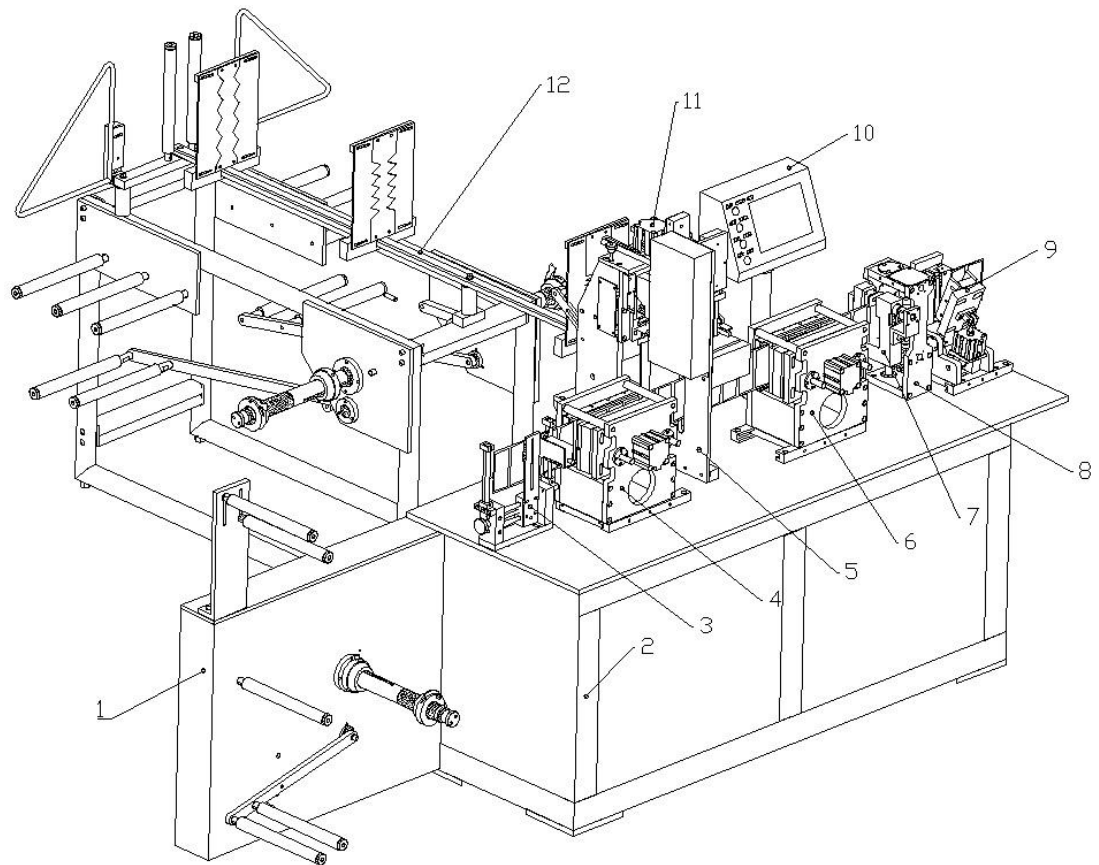
2.2 Main Technical Paramete

Pouch size	Length 40--130MM, Width 40--80MM (it can be made as requirement)
Package material	PP, PE, compound film, aluminum foil film, paper film, etc.
Diameter of inside roller of package film	Φ 70-75mm
Diameter of package film	Maximun Φ 600mm
Width of package film	80-250mm
Wet tissue material	0-80g/m ² air laid paper, nonwoven, spunlace nonwoven fabric etc

Diameter of inside roller of tissue material	ϕ 70-75mm
Diameter of tissue material	Maximum ϕ 1200mm
Width of tissue material	Maximum ϕ 260mm
Folded type	Max 8 folding vertical, and 4 folding horizontal
Production speed	60-120 bags/min(according to the size and component of wet tissue)
Liquid added range	0-10ml
Electric power	Single phase ,220V ,50 Hz
Working air condition	0.6-1.0 Mpa
Air consumption	$\geq 0.2\text{m}^3$ /min(w-1.0/10 air compression machine is needed)
Machine size(L*W*H)	2800X2800X1600mm
Package size of the machine(L*W*H)	3000X1000X1800mm

Net weight	900kg
Gross weight	1000kg

2.3 Structure



1 package film install unit 2 rackmount

3 package bag forming unit

4 bottom sealing part

5 tissue forming and liquid adding part

6 sealing part

7 easy tear part shaping

8 product feeding part

9 cutting part

10 PLC and touch screen controller

11 tissue cutting part

12 tissue folding part

3. Installation of machine

3.1 Requirement of floor

The machine should be installed on the flat and firm ground, the enough operation and maintenance space should be kept around the machine.

3.2 Environment condition

- 1) Temperature: the normal environment temperature should be 20°C~28°C.
- 2) Humidity: 50%~70%.
- 3) Keeping away from oil, gas, chemical matters, acid, alkali and explosive or the inflammable.
- 4) intensity of illumination: more than 300LUX.
- 5) Avoid direct sunlight.
- 6) avoiding closeing to the machine bringing shakes.
- 7) Avoiding the fan directly blowing.
- 8) Avoid disturbance of strong signal

3.3 Test of electricity supply

1) 220V, 50HZ is suitable for this machine, 380V or other suitable power supply is forbidden.

2) Ground wire is required. The cross section of copper is not less than 2mm² and the cross section of aluminum is not less than 4 mm². And the ground wire must be connected with "PE" part of the machine.

3) Connect the machine with the air compressor before it works. And the diameter for pipe should be 12mm, air pressure should be between 0.6 and 1.0 Mpa, air filter is added with turbine oil.

4) Switch off different breakers in control box to perceive whether various meters on console and electricity power signal is normal or not.

- 5) press the point-touch button to notice whether the working condition is right.
- 6) Enter 【自动页】 page, and set 40mm/s for the speed of feed film , and set 100mm/s for the speed of feed tissue. Then press start bottom (green bottom) or press 【run 】 to make the machine work for 3 or 5 minutes. If it works well, then press stop bottom (yellow bottom) or press 【stop】. If the machine does not work well ,press the urgent stop bottom.

4. Machine setting and test

4.1. Operation instruction for control panel

The control panel of the machine consists of URGENT, POWER, START, STOP button and the touch screen.



URGENT: Press this button to stop the running equipment and cut off the power supply.

POWER: Press this button to connect the control loop power supply; the PLC touch screen is lighted and enters **[HOME]**.

START: Press this button to start the machine. The operation is same to clicking the **START** in **[AUTO]** on the touch screen.

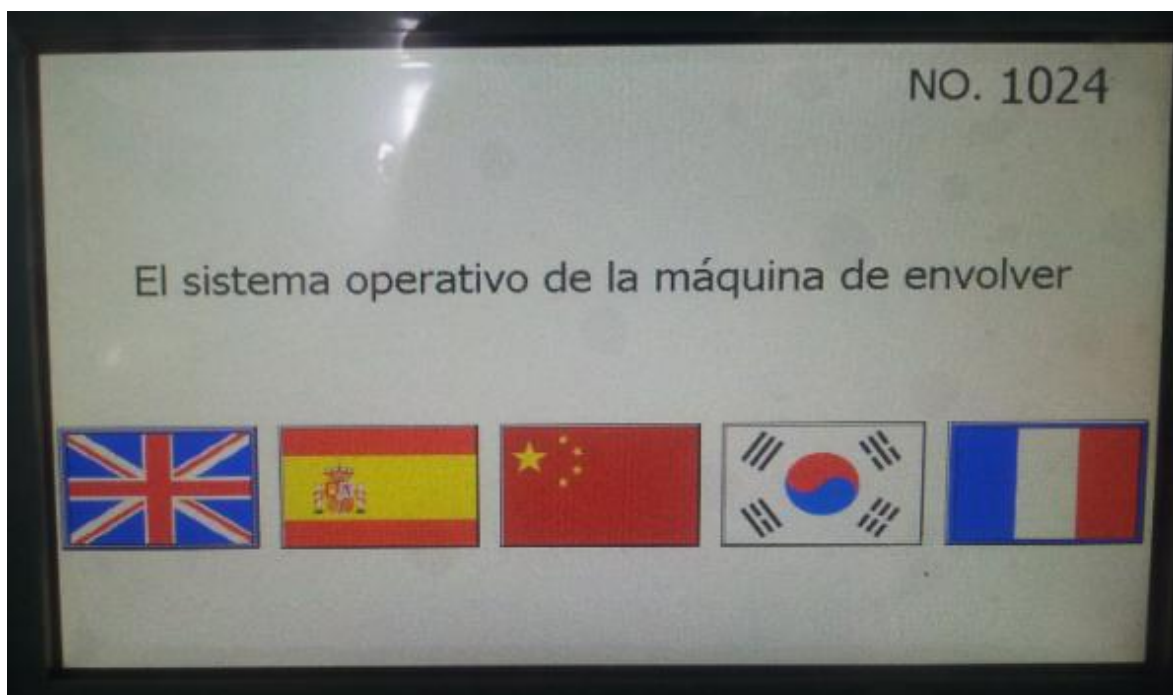
STOP: Press this button to stop the machine. The operation is same to clicking the **STOP** in **[AUTO]** on the touch screen.

Touch screen: Control the running and stop of the entire machine, configure the parameters of the machine and display the running state of the machine.

4.2. Instructions for PLC operation

VPD-350 automatic horizontal four sides sealing wet wipes packing machine is controlled by PLC touch screen. Avoid scratching the touch screen with sharp object or damage due to overheating. (The **[AAA]** in the manual is the AAA page of the operating system, and **BBB** is the BBB button. For example, click **START** in **[AUTO]** to start the machine, it means to click the “**START**” button in the “**AUTO**” page of the operating system to start the machine.)

1) **[HOME]**



When the power supply of the touch screen is connected, it enters the **[HOME]** page automatically, which consists of company name, operating system version of the packing machine, contact, company website and **ENTER** key.

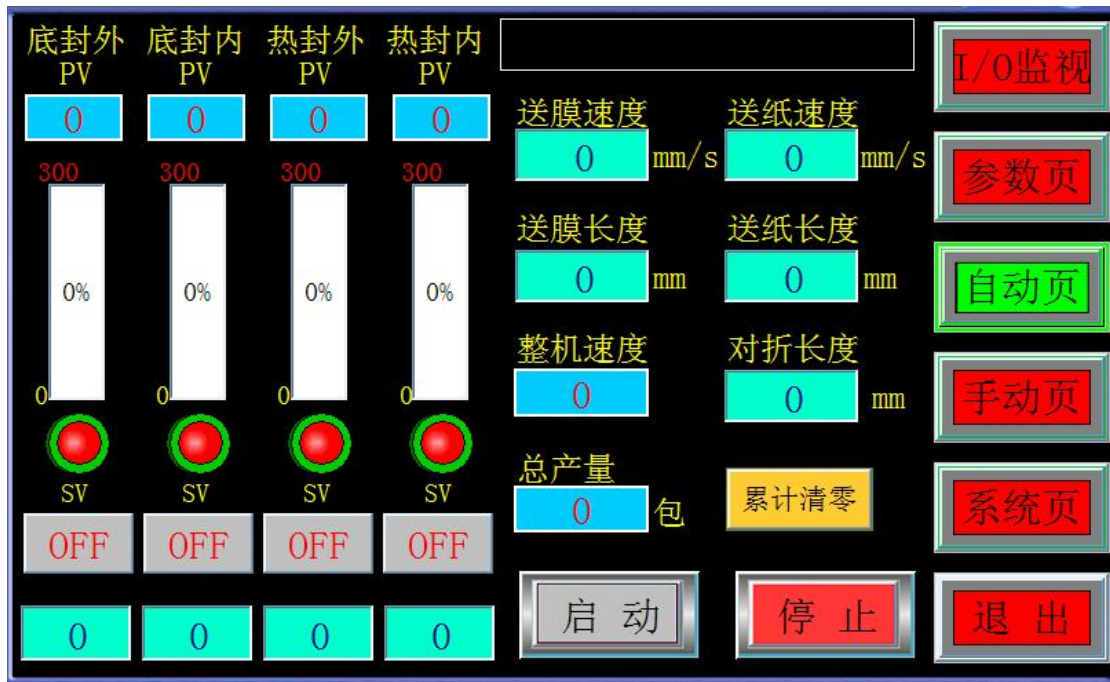


ENTER: Click **ENTER** to enter **[AUTO]** page.



In **[IO MONITOR]**, **[PARAMETERS]**, **[AUTO]**, **[MANUAL]** and **[SYSTEM]** pages, they consist of the page tab bar in the left and the current page content in the right. The page tab bar is shown in the left.

The current tab in the tab bar is in green, indicating that the content in the left is current page. To switch to next page, click the tab directly. Click the **“EXIT”** tab to return to **[HOME]**.



底封外 PV 底封内 PV 热封外 PV 热封内 PV

Outside bottom sealing/ Inside bottom sealing/ Outside hot sealing/ Inside hot sealing

送膜速度 送纸速度

Film feeding speed/ Tissue feeding speed

送膜长度 送纸长度

Film feeding length/ Tissue feeding length

整机速度 对折长度

Machine speed/ Folding length

总产量 0 包 累计清零

Total output

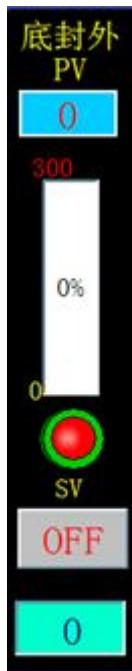
Bags

Reset

启动 停止

[AUTO] consists of temperature control and running control.

The temperature control consists of outside bottom sealing, inside bottom sealing, outside cushion seal and inside hot-seal. Each row is one group (as shown in the figure in the left).



First line is the name of the group, such as outside bottom sealing;

Second line is the actual temperature of the corresponding module after heating (unit: °C);

Third line displays the actual temperature in red bar. High red bar indicates high temperature, low red bar indicates low temperature and the empty bar indicates the room temperature.

The fourth line is a hollow circle. Red in the circle indicates that the power supply is cut off, the module isn't heated or the heating pauses; green in the circle indicates that the power supply is connected and the module is being heated.

The Fifth line is the heating switch. The red OFF on the switch indicates that the heating switch is disconnected; click the switch to display ON and the characters turn into blue, indicating that the switch is connected and being heated; click the switch again to disconnect.

The Sixth line is the heating temperature that has been set. After heating, if the actual temperature reaches the set value, it stops heating automatically; when the actual temperature is lower than the set temperature, it resumes heating automatically. Click this data to have the following dialog box:

数值型:

0							
1	2	3	4	5	.	<—	确定
6	7	8	9	0	--	CE	取消

Numerical type:

The dialog box in the figure above is the **numerical keypad**. In the operating system

of VDP-350 packing machine, this keypad pops up when you need to enter data. The second line is the figure you have entered. The numerical keypad consists of ten keys (0-9) and radix point. You can press "--" to enter "-", press "<-" to clear previous figure, press "CE" to clear all figures, press "OK" to confirm the enter and replace the original data with the figure in the second line, and press "Cancel" to exit and reserve original data.

The running control consists of the following items. The items that allow parameter settings are displayed in light blue. Click the data and set the parameters with the popup keypad.



Display the faults diagnosed by the system automatically for maintenance.



This is the speed that the packing film feeding mechanism feeds the packing film.



This is the length that the packing film feeding mechanism feeds packing film every action, which is set according to the packing width.



This is the speed that the tissue cutting mechanism feeds the tissue.



This is the length of the tissue after cut by the tissue cutting mechanism and is set according to the actual length of the tissue.



If the fold length is 0, the tissue cutting mechanism feeds the required tissue in one time and then cuts it off.

If the fold length is larger than 0, the previous feeding length by the tissue cutting mechanism is the fold length; after the "Fold stay" time, the machine feeds the remaining length and cuts it off. That is to say,

one piece of tissue is fed in two times: feeds the fold length first, and then feeds the remaining length after the “Fold stay” time. The “Fold stay” time can be set in Fold stay in [PARAMETERS].

The fold length is a figure between 0 and the feeding length and can be set accordingly.



Display the punching times every minute of the cutting mechanism when the machine is running.



The accumulated punching times of the cutting mechanism since last clearing.

Bags



CLEAR: Clear the counting and start new counting.

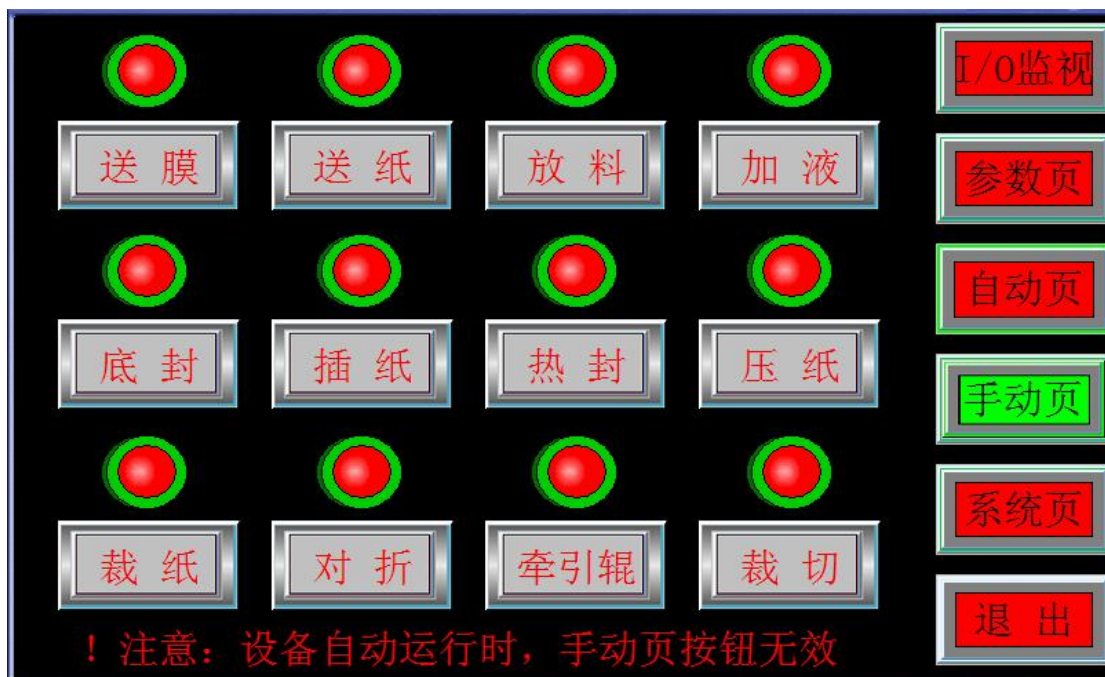


START: Start the machine and produce according to set parameters.



STOP: Stop the running of the machine.

3) [MANUAL]



Note: When the machine is in auto running mode, all the keys in [MANUAL] page are invalid.

As shown in the figure above, the manual page is mainly used for machine test and every button corresponds to the mechanism. Except the traction roller, all the other buttons are clicking buttons. When you click a button, the dot turns to green from red and the corresponding mechanism acts; after releasing, the dot resumes red and the corresponding mechanism resets. If you click the traction roller, the dot turns to green from red and the roller clamps; click it again, the dot resumes red and the roller loosens.

Note: When the machine is in auto running mode, all the keys in [MANUAL] page are invalid.

4) [PARAMETERS]



The parameters page is used to set the action time and other parameters of the main mechanism when the machine is in auto running mode. The set data are displayed in light blue and you can click the data to pop up the numerical keypad. The details of action time setting is shown below:

Film feeding coefficient: Modify the error between set film feeding length and actual

film feeding length.

Fold stay: If the fold length is larger than 0, it is the stay time after feeding the fold length

Liquid filling delay: Set the time (unit: second) that the liquid filling mechanism acts downwards before the metering device starting liquid filling metering

Tissue feeding coefficient: Modify the error between set feeding length and actual feeding length

Cutting times: Set the times that the packing film feeding mechanism acts before the cutting mechanism cutting once

Tissue pressing delay: Set the time that the liquid filling mechanism acts downwards before the tissue pressing mechanism starting pressing

Color code quantity: The quantity of packing film that the feeding machine feeds in one time in tracking mode wet wipes packing

Tissue pressing time: Set the time that the tissue pressing mechanism presses the tissue before releasing the mechanism

Bottom sealing time: Set the time that the bottom sealing mechanism presses before releasing the mechanism

Fold time: Set the time before the fold arm resetting

Tissue inserting time: Set the time before the tissue inserting board inserting tissue

Hot-seal time: Set the time that the hot-seal mechanism presses before releasing the hot-seal mechanism

Cutting time: Set the time that the cutting mechanism acts before releasing the cutting mechanism

5) [IO MONITOR]

IO MONITOR is mainly used to monitor the working state of I/O points of PLC module,so as to observe the working state of the mechanisms and the state of inspection points

The items starting with I are input items, the items starting with Q are output items, while those have no characters after the items are blank temporarily

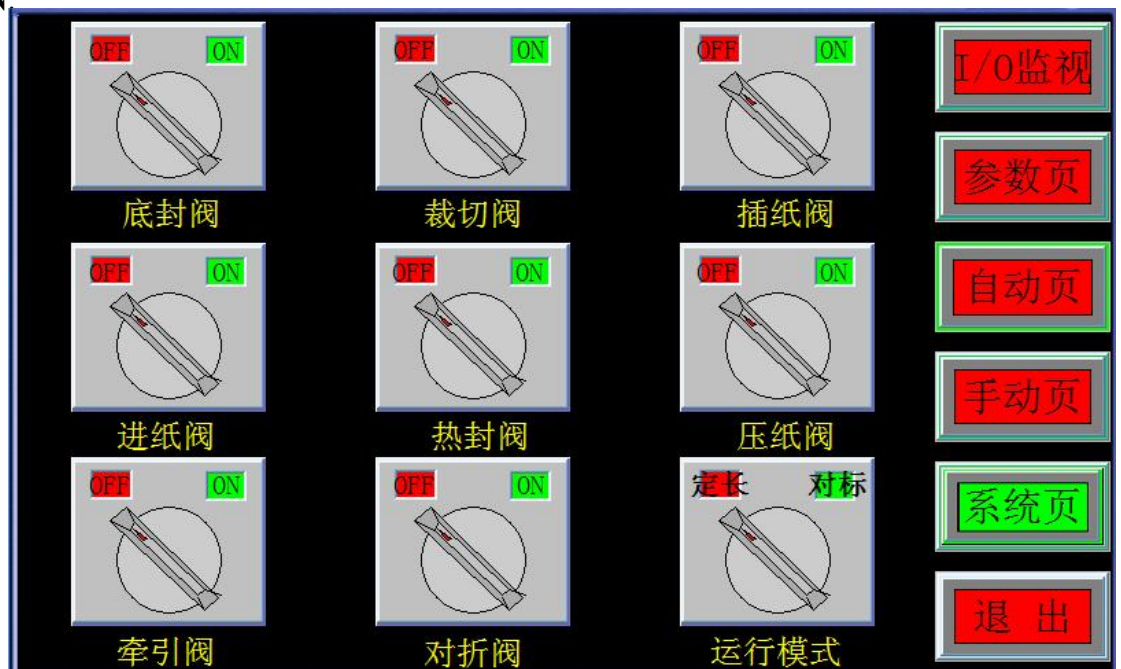
If the dot before an item is red, this input point or output point doesn't have signal input

or output
If the dot
output



6) [SYSTEM]

The **SYSTEM** page is used to enable or disable the mechanisms such as bottom sealing, tissue inserting, hot-seal, traction, cutting, tissue feeding and tissue pressing when the machine is in auto running state. If **ON** is selected, this mechanism acts when the machine is in auto running mode; if **OFF** is selected, this mechanism doesn't act when the machine is in auto running mode. It also includes the selection of packing mode. Two packing modes are available: tracking mode, which packs the wet wipes in printed patterns; fixed length mode, which packs the wet wipes according to specified film feeding length. You can click the switch to select **ON** or **OFF**, Tracking or Fixed length. The function is disabled if the switch points to **OFF**, and enabled if it points to **ON**.



Note: The operating system of VDP-350 four sides sealing horizontal wet wipes packing machine has different versions and thus the PLC operation instructions in this manual may be different from the actual machine. Please contact us for details.

4.3. Temperature control

This machine controls the “outside bottom sealing”, “inside bottom sealing”, “outside hot-seal” and “inside hot-seal” with four temperature control systems. The hot-seal temperature is 100—160°C generally. The optimal temperature during working depends on sealing pressure, sealing time, packing material and ambient temperature. Please adjust according to the sealing. The optimal value shall be able to prevent leaking and coke wrinkle in the hot-seal position. The pre-heating time of hot-seal is 10-20 minutes generally.



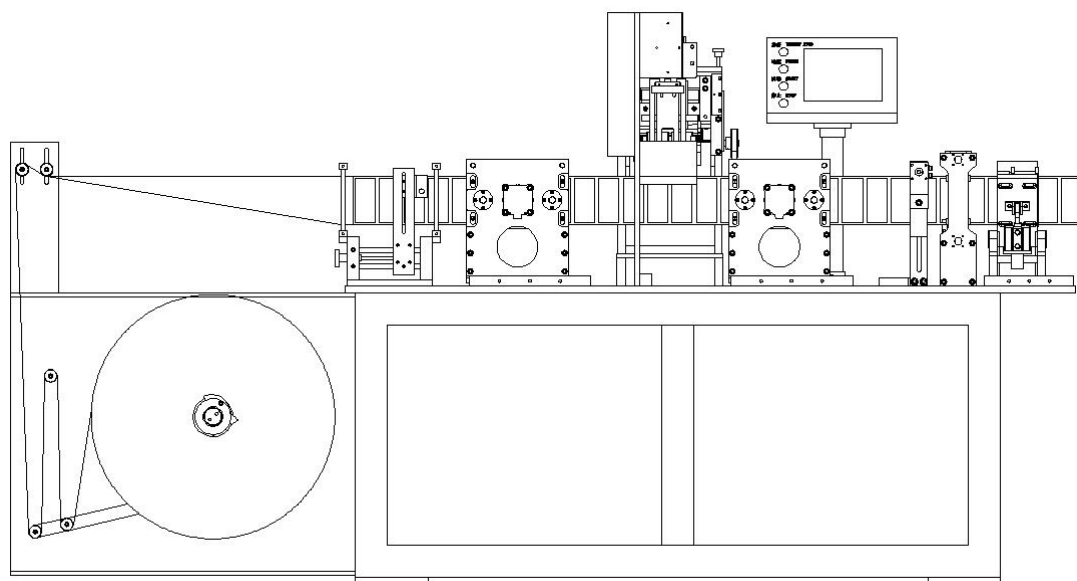
Enter [AUTO] page in temperature setting:

Click the data in the blue box in the bottom line of every group to pop up the numerical keypad and set the desired temperature value. Click the heating switch in the fifth line in every group to switch to ON. If the dot in the fourth line turns to green

from red, the heating switch is connected and the module starts heating. When the module temperature rises to specified value, the dot turns to red from green, indicating that the automatic heating is disconnected and the module stops heating; when the module temperature drops to specified value, the dot turns to green from red, indicating that the heating is connected automatically and the module resumes heating; repeating in this way, the module temperature is kept at specified value.

4.4. Installing packing film

Install the packing film on the reel and keep the exit of the packing film to the left in the top, adjust the tissue lock wheel to keep the tissue film in the center of the roller (i.e. the center position of bag forming mechanism). For the midline error of the packing film after folding, adjust slightly since the effect won't respond immediately. The rounding path of the packing film is shown in the figure below. When the packing film is installed on the reel, it is necessary to wind in certain path and pass through bag forming mechanism, bottom sealing, tissue inserting liquid filling mechanism, hot-seal, easy tearing punching, packing film feeding mechanism and cutting mechanism in sequence. Thus the film is installed.



4.5. Packing in fixed length mode

VDP-250 packs wet wipes in two modes: fixed length mode and color code tracking

mode

Fixed length mode: It is suitable for the packing of wet wipes that have no printed patterns on the surfaces. The machine cuts the packing film according to the value set in the **Film feeding length** in [AUTO] page of PLC.

First, select the fixed length mode in **Mode** in [SYSTEM] page of PLC, and then set the required packing width in **Film feeding length** in [AUTO] page. The value of bag width should be consistent to the sealing and cutting module. When the packing film is folded, it forms the length of the bag automatically. After the bottom sealing, the packing film is made into bags primarily. Please check and adjust in the following sequence:

- 1) The top edge of the packing film in tissue inserting liquid filling mechanism is in the bottom of the tissue holder; if not, please adjust the position of packing film.
- 2) Check whether the sealing impression width of the bottom meets the requirement after sealing; if not, adjust the position of bottom sealing module.
- 3) When the tissue inserting board is inserted into the bag, the board should be in the center of the bag; if not, adjust the position of the bottom sealing mechanism.
- 4) When the tissue inserting board is inserted into the bag, the bottom edge of the board should be 4-8mm away from the top line of the bottom seal; if not, adjust the position of tissue inserting board.
- 5) The sealing impression on both sides after hot-sealing should coincide with the sealing impression of the bottom seal; if not, adjust the position of the hot-seal mechanism.
- 6) Check whether the sealing impression width of the bag opening after hot-sealing meets the requirement; if not, adjust the position of hot-sealing module.
- 7) Check whether the cutting position is in the center of the sealing impression; if not, adjust the position of the cutting mechanism.
- 8) Check whether the V opening is on the edge of the bag after cutting and whether the top position and bottom position meet the requirement; if not, adjust the position of the punching mechanism.

4.6. Packing in color code tracking mode

Color code tracking mode: Suitable for the packing of wet wipes with printed patterns on the surfaces. The machine tracks the “color code” printed on the packing film through photoelectric eye and calculates the accurate bag length automatically to cut the packing film.

The function of photoelectric eye color code tracking system is to monitor the length of the packing film used by every bag (the packing film that isn't printed with color code can't be monitored; the distance between color codes printed on the packing film should be consistent to the size of sealing and cutting module). The color code printed on the packing film should be contrasty to the color around. The size of printed color code is 8-10mm in cross direction and 3-5mm in vertical direction.

First, select the tracking mode in **Mode** in **[SYSTEM]** page of PLC, and then adjust the position of photoelectric eye, making the light beam from the color code photoelectric eye aim at the position that the color code passes, and keep the distance between the surface and packing material to 8-12mm. When the color code point on the packing material passes through photoelectric eye, the red indicator and green indicator on the photoelectric eye shall flash alternatively. At the same time, the **[IO MONITOR] X0 – color code photoelectric** on the PLC panel flash alternatively.

After the bottom sealing, the packing film is made into bags primarily. Adjust according to the eight steps in fixed length mode, and then adjust the position of the tracking photoelectric eye, making the “color code” in the center of the sealing impression of the two bags.

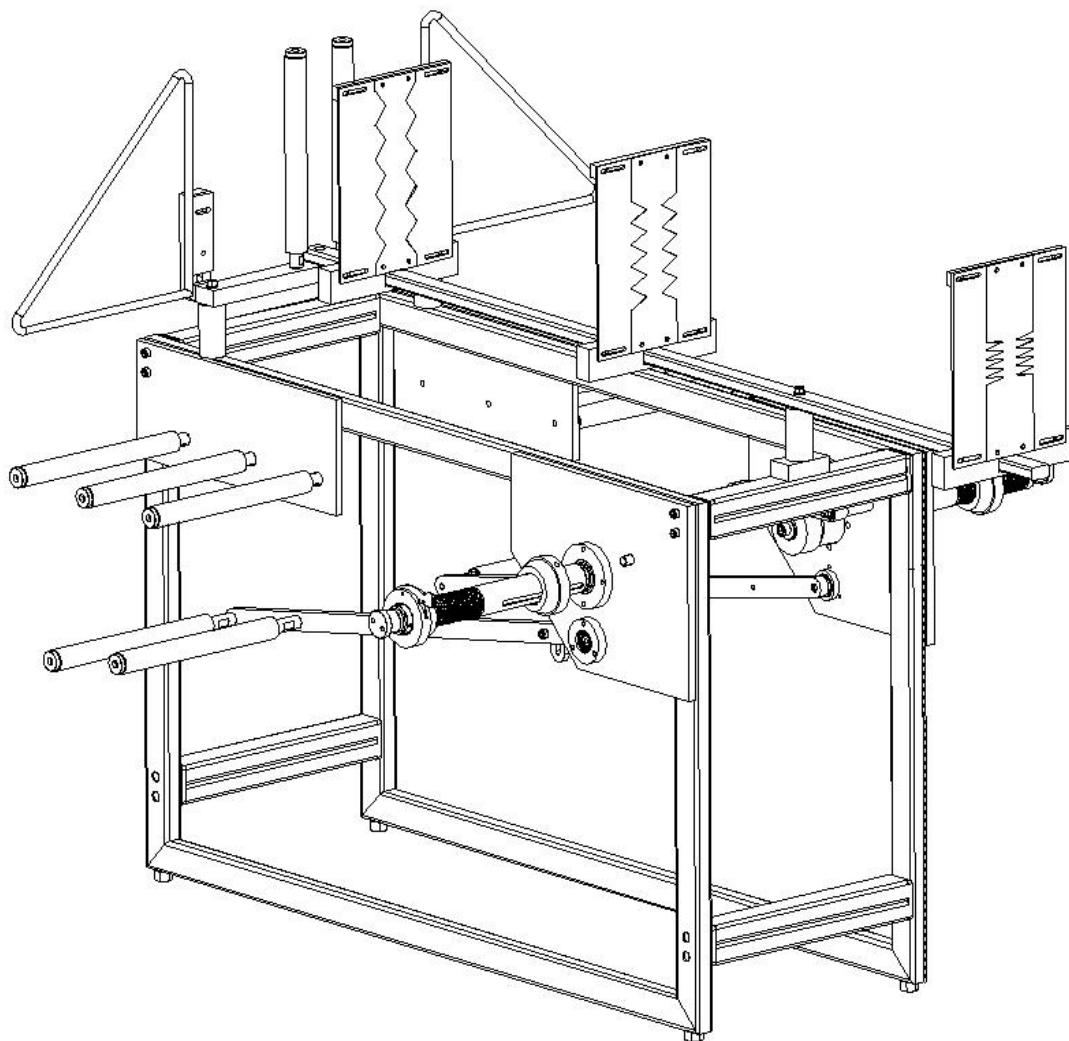
4.7. Adjusting packing speed:

The production of VPD-350 wet wipes packing machine can't be adjusted directly. You can adjust the film feeding speed, tissue feeding speed, bottom sealing time, tissue inserting time, hot sealing time, cutting time, tissue cutting, and thus change the packing speed of the machine. The film feeding speed and tissue feeding speed should be adjusted together with bag width and tissue length to ensure the coordinate action of every mechanism. Refer to **[AUTO]** page and **[PARAMETER]**

page for details. The PLC [AUTO] page contains packing speed display and automatic counting.

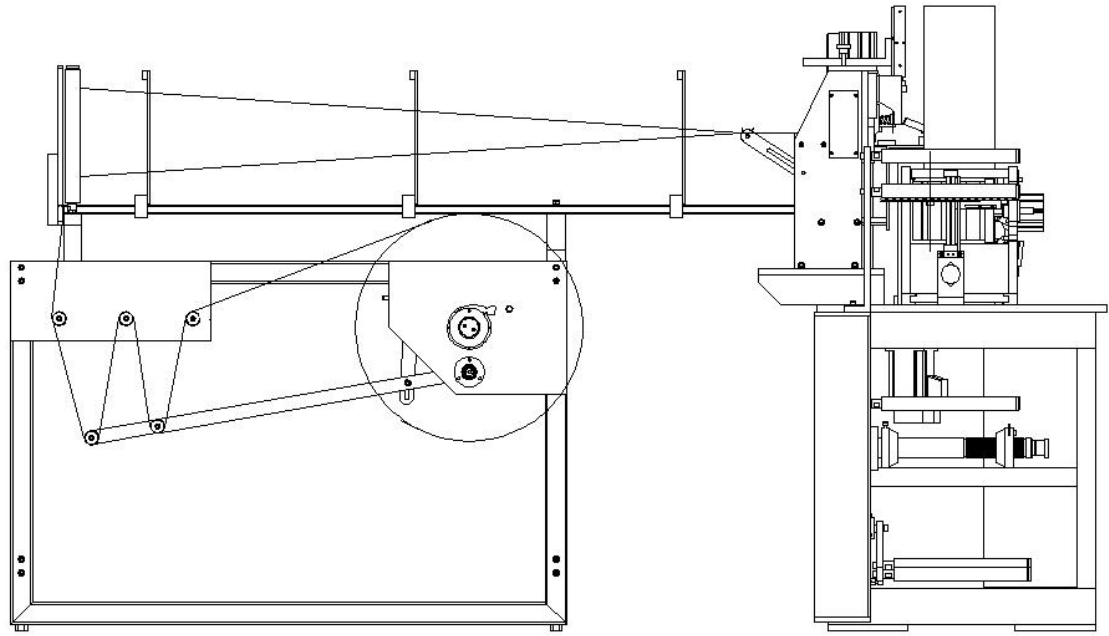
4.8. Tissue unwinding folding mechanism

To make the packing and transportation more convenient, the tissue unwinding mechanism should be removable. This mechanism integrates three support pillars (diameter: 40mm, length: 80mm). The support pillars are dedicated for third fold and fourth fold, that is to say, the support pillars are used when the wet wipes are folded for three or four times, and can't be used when the wet wipes are folded in cross direction. The assembly of the mechanism is simple and easy, and the effect after assembling is shown below:



- 1) The tissue is installed on the reel, keep the exit to the left in the bottom as

shown in the figure below, and pass through the fold roller, reel roller, turning device, fold former and enter tissue cutting mechanism traction roller.



2) The tissue is folded in 2-8 times vertically through fold forming machine, which determines the fold width. Adjust the hackle gap of the fold forming machine for trimming. The tissue is pressed into strips by the vertical roller and forms an isosceles triangle; if not, move the position of the fold forming machine. If the tissue isn't folded symmetrically, please adjust the hand wheel and move the position of the scroll, as shown in the figure above.

3) The tissue length and feeding speed are set in [AUTO] page of PLC.

4) When the tissue length is changed, move the tissue cutting mechanism according to the actual condition, to ensure that the tissue inserting liquid filling mechanism is in correct position while folding after the tissue is fed to the tissue holder.

5. Machine operation

5.1. Manual running

Click the desired button in [MANUAL] page of PLC, and the dot in the top

turns to green from red and the mechanism acts; after releasing, the dot resumes red and the mechanism resets. After clicking traction roller, the dot in the top turns to green from red and the traction roller clamps; click the button again, the dot resumes red and the traction roller releases. Click the buttons in sequence to complete the packing of wet wipes.

When the machine is in automatic running mode, all the keys in [MANUAL] page are invalid.

5.2. Automatic running

- 1) Set the film feeding speed, film feeding length, tissue feeding speed, tissue feeding length, and select fixed length or tracking mode;
- 2) Adjust the position of every mechanism according to the selected packing mode for wet wipes
- 3) Click the **START** (green) button and the machine starts continuous running in set parameters until the **STOP** (yellow) or **URGENT** button is pressed.

5.3. Stop

- 1) Normal stop: Click the **STOP** (yellow) button and the machine stops running immediately.
- 2) **Urgent stop:** Press the **URGENT** button in any direction to shut down the machine immediately.

5.4. Operating steps

- 1) Before turning on the machine, lubricate the positions that have relative motion.
- 2) Check whether there are sundries on the sealing module, worktable and cutting mechanism, and make sure that no person is operating the machine.
- 3) Turn on the power switch, and check whether the power signal, human-machine interface and control panel are normal.
- 4) Set the temperature of bottom sealing and hot sealing modules.
- 5) Install the packing film on the machine following the winding path.
- 6) Install the tissue on the machine along the winding path, and adjust the position of fold forming machine properly.

7) Set the film feeding speed, film feeding length, tissue feeding speed and tissue feeding length, select fixed length mode or tracking mode, switch off the power supply of liquid filling pump temporarily and disable tissue feeding.

8) Adjust the position of every mechanism according to selected wet wipes packing mode

9) Start the machine and check the folding of the packing film.

10) Check whether the bottom sealing, tissue inserting, hot sealing and cutting inosculate again.

11) Check whether the sealing is firm. If the temperature is too low, the sealing won't be firm; if the temperature is too high, the surface will be in white and cause failure easily.

12) Check the cutting, which should be able to cut off the tissue and won't make loud impact noise.

13) Check whether the photoelectric eye irradiates on the orientation color code of the packing material. When the color code passes, the red indicator and green indicator on the photoelectric eye should flash alternatively, and the dot on PLC **[IO MONITOR]** X0 – color code photoelectric eye flashes. If not, please adjust in previous method.

14) Stop the machine, feed tissue manually, cut and check whether the cutting of wet wipes is normal.

15) Check whether the position of the wet wipes is proper; if not, move the position of tissue cutting mechanism until it is proper.

16) Turn on the power supply of metering liquid filling pump, insert tissue manually and check the tissue inserted in the bag.

17) Check whether the liquid filling of wet wipes meets the requirement.

18) When the above steps are finished, turn on the machine and start packing.

6. Maintenance and cleaning of the machine

6.1 External cleaning of the machine.

External cleaning is mainly to clean the table-board, outline, scraps of packaged

objects. Clean the machine with dry or twisted wet cloth. Cleaning with water or wet washing is prohibited. (If the machine housing is sprayed with plastics, sterilizing or wiping the machine with alcohol or any volatile liquid is not appropriate. Otherwise, it may result in outside discoloring.)

Make sure the primary power supply is cut off to maintain the human-machine safety before cleaning and wiping the machine.

6.2 Inside cleaning of the machine

The components to be regularly cleaned by hand are as follows:

1. Regularly clean up every electrical component in the electrical cabinet by using a soft hairbrush or air blowing to keep all of them clean and well-functioned. Check whether the exhaust blower is in order or not.
2. Clean the groove transducer inside the machine with a hairbrush. The fluted disc side of the gearbox can only be dry cleaned.
3. Clean the cumulative dirtiness of the carbon brush on the collector ring to keep normal power-on.
4. While cleaning, make sure to cut off the primary power supply and ensure the human-machine safety.

6.3. To lubricate the machine

For the machine is not equipped with automatic oiling mechanism, drive components like jogged gear cluster and sprocket-wheel cluster shall be regularly lubricated by hand so as to prolong the lifespan of the machine. It's recommended to adopt butter and mechanical oil.

It's better to lubricate the machine in light lubrication and slight oiling. Oiling too much may result in splash of oil while components are turning. Notice that all kinds of belts including conveyer belt mustn't be lubricated or oiled while oiling.

For the lubrication of model P toothed-chain stepless gearbox, refer to the Instruction Manual and Maintenance. Relative technical documents shall be regularly checked and cannot be neglected.

7.General trouble shooting

Contents	Causes	Solutions
Incapable to cut the packaging bag off.	1. Insufficient spring force of the cutter. 2. Insufficient temperature of the cutter. 3. Abrasion of the cutter	1. Readjust the pressure. 2. Raise the horizontal-sealing temperature slightly (refer to P9 and P12) 3. Readjust or replace.

Much noise arising from the engagement of horizontal-sealing cutter	1. Impact arising from the engagement of the upper cutter and the lower one. 2. The tooth spaces of the upper cutter bed and the lower one are out of position .	1. Reduce the number of the copper sheets of the lower cutter. 2. Refer to the Items 6 and 13 in P12 for adjustment.
The packaged contents exposed to cut	1. Incorrect setting for the height of the crosscut cutter. 2. Too much remained air in the empty bag. 3. Improper feeding of feeding chain hook. 4. Too fast package speed. 5. Too loose packaging bag.	1. Readjust height. 2. Adjust the flattening hairbrush to eliminate air. 3. Readjust the position of the feeding hook. 4. Lower the package speed. 5. Tighten the bag maker appropriately.
Two sheets of films are unaligned when the packaging bag is vertically sealed.	1. The winding drum of the film deviates from the feeding center. 2. The packaging bag deviates from the feeding center.	1. Adjust the inside and outside positions of the winding drum. 2. Adjust and modify the flattening hairbrush.
Obscure printing	1. The printer can not start. 2. Insufficient touch between the steel character and the film. 3. The printing ink is used up. 4. Too low set temperature.	1. Adjust the friction wheel. 2. Adjust the pressure between the inker and the steel character. 3. Replace the inker. 4. Raise the temperature slightly.

The bag length doesn't reach the designated position after setting the length.	1. Groove transducer is dirty or damaged. 2. Travel switch of the gearbox is damaged. 3. No speed change for the gearbox. 4. Limit travel switch for setting paper length shifts 5. The motor for adjusting the bag length does not work or run in one way.	1. Clean both sides of the groove or replace it. 2. Replace travel switch. 3. Repair gearbox. 4. Reset and readjust limit switch. 5. Check the circuit or replace the motor.
--	---	--

No display of the bag length	1. Damaged groove switch of the gear case or too dirty at both sides. 2. Inaccuracy positions of the fluted disc and the groove switch. 3. Damaged proximity switch of the cutter bed	1. Clean or replace. 2. Readjust. 3. Replace proximity switch.
Compensation red/green light is not on.	1. Red light is on but green light off, or green light is on but red light off. 2. The light-accuracy light is off.	1. Reset the length by increasing or decreasing. 2. Check the circuit, and repair or replace indicator lights.
Temperature controller can not control the temperature or display.	1. Damaged heater pipe. 2. Damaged solid-state relay. 3. Damaged thermoelectric couple. 4. Burnout of temperature controller	1. Replace. 2. Replace. 3. Replace. 4. Replace.
The temperature controller leaps.	1. Too dirty diversion copper collar. 2. Poor contact between the copper collar and the brush. 3. Malfunction of the thermoelectric couple.	1. Clean copper collar. 2. Readjust or replace the block brush. 3. Replace.

Air leakage or breach of the vertical-sealing	1. The temperature is on the high side or low side. 2. Too loose or too tight rolligon wheels 3. Poor packing paper film	1. Adjust the temperature to an appropriate value. 2. Adjust rolligon wheels. 3. Replace paper film.
---	--	--

Air leakage or worn-out film of the horizontal-sealing	1. The temperature is on the high side or low side. 2. Dirty cutter block. 3. Misplacement between the upper and lower cutters. 4. Poor-quality packing paper film.	1. Adjust the temperature to an appropriate value. 2. Clean up with a copper-wire brush. 3. Refer to Items 6 and 13 in P12. 4. Replace paper film.
Horizontal-sealing cutter cuts behind the color code even with the deviation constantly cumulated	Too long set length of packaging bag.	Reset the length of packaging bag by decreasing the value (about 1 mm)
Horizontal-sealing cutter cuts at the front of the color code even with the deviation constantly cumulated	Too short set length of packaging bag.	Reset the length of packaging bag by increasing the value (about 1 mm)

Contents	Causes	Solutions
Paper throw	1. No alignment with the paper film. 2. The tension of paper film is not well adjusted. 3. The former is too lower and the feeding hook contacts paper. 4. The former clamps too narrowly.	1. Adjust the load roller handwheel by turning clockwise or anticlockwise to determine the centre. 2. Refer to P7 and P8. 3. Raise the former high without contacting the paper film. 4. Release the forming plate appropriately.
The deviation rectifying	1. Reversible motor works in unidirection or motor doesn't	1. Check whether PLC signals is in order and shoot the circuit

device doesn't work.	work. 2. Damaged start capacitor. 3. Damaged relay. 4. Inaccurate planetary differential gear. 5. Damaged solid-state relay.	trouble. 2. Replace the capacitor. 3. Replace the relay. 4. Adjust or replace the parts and bearings. 5. Replace.
Shift of the tracing photoelectric eye	1. Incorrect length. 2. Dirty paper-feeding tracing roller. 3. Non-response of the photoelectric eye. 4. Non-response or deviation of the proximity switch for horizontal-sealing cutter. 5. Too tight or loose brake 6. Improper compensation amount.	1. Reset the length referring to Items 5 and 6 in P7. 2. Clean up the roller. 3. Reset sensitivity according to the instruction to the photoelectric eye. 4. Keep the distance between the proximity switch and the jogged wheel for 2-3 mm. The proximity switch shall be aligned with the position block on jogged wheel when the upper and lower cutters are in position. In this moment, the light shall be on. 5. Adjust the tension of brake spring. 6. Reset the compensation amount by increasing or decreasing.

VIR00 PACKING MACHINERY

VPD-250 APPARATUS LIST

序号	名称	产地	型号及规格	单位 pcs	数量
01	主机 Machine cover		Stainless steel 外壳不锈钢	台 pcs	1
02	触摸屏 touch screen	WEINVIEW	MT506MV	只 pcs	1
03	伺服电机 servo motor	松下 Japan panasoni c	MSMD082PIU	只 pcs	1
04	伺服电机 servo motor	松下 Japan panasoni c	MSMA102PI G	只 pcs	1
05	模块 block	松下 Japan panasoni c	FPO-C16T	个 pcs	2
06	电源开关	台湾明伟	D-60B	只	1

	power switch	taiwan		pcs	
07	固态接触器 solid contactor	上海 shanghai	SSR-25DA	只 pcs	2
08	加热模块 heat block	金凌 jingling	HR-TCI-A	只	1
09	按钮 botton	日本 Japan	欧姆龙	只	4
10	电磁阀 electromagne t	台湾 taiwan	220V	只	7
11	气缸 cylinder	经纬 jingwei		只	10
12	接进开关 switch	上海 shanghai	LJ12A3-4-ZB X	只	3
13	光电眼跟踪 photo electrical testing transducer	上海亚泰 shanghai	BZJ-211	只	1
14	气流减压阀 air decompress valve	上海 shanghai		只	9

15	计量泵 metering pump	日本 Japan	IWAKI	只	2
-----------	----------------------------------	---------------------	--------------	----------	----------